Loan Approval Prediction using Decision Tree

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Project & Data Overview

Predicting Personal Loan Acceptance

1

Project Objective

The project aims to predict whether a customer will accept a personal loan offer using a Decision Tree model. This enables precise marketing, reduces acquisition costs, and improves campaign ROI.

2

Data Source

We use the Universal Bank dataset, which contains 5,000 complete customer records with 14 attributes, including age, income, family size, education, mortgage, and product-usage flags.

3

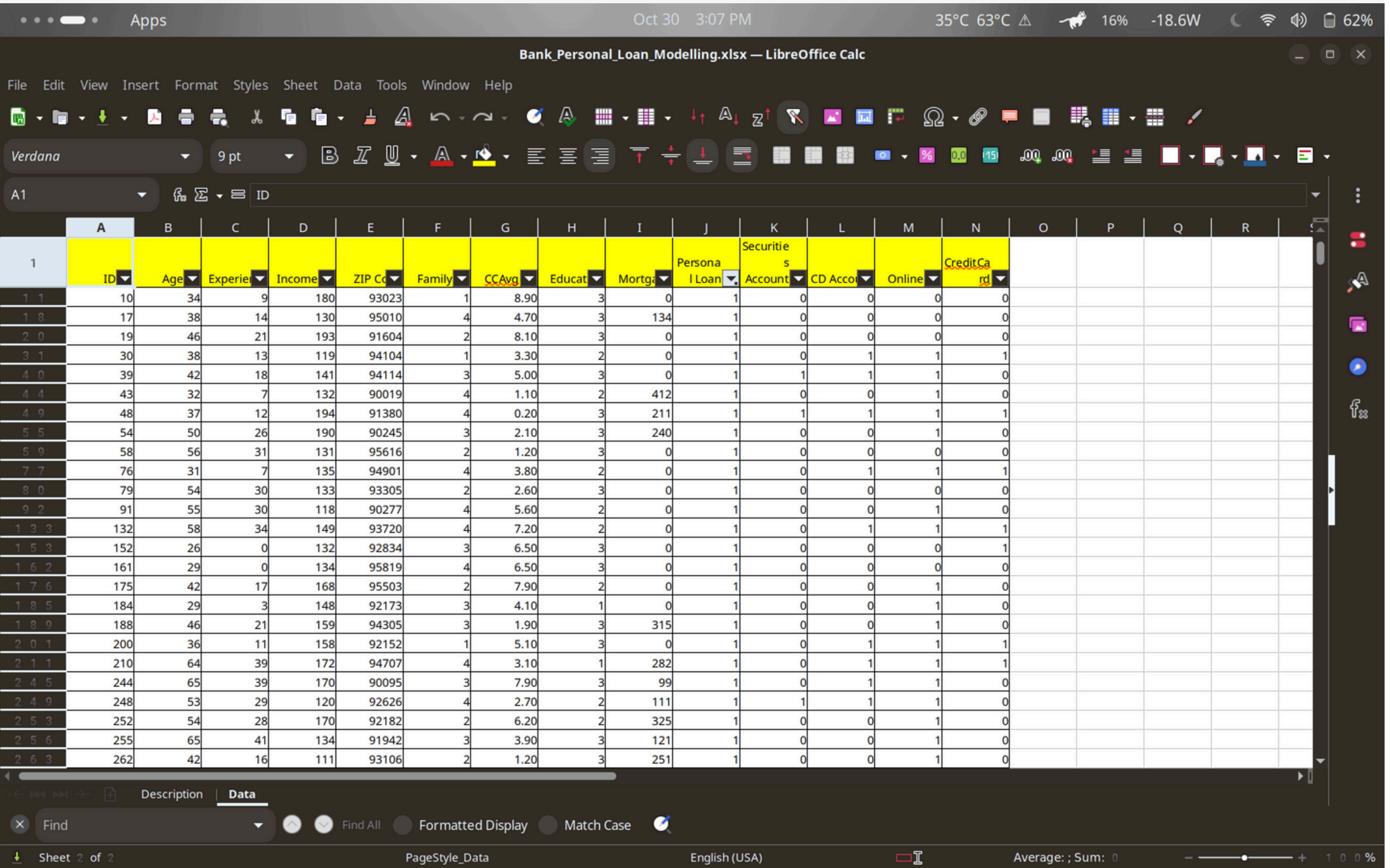
Supervised Classification

The dataset provides a solid foundation for supervised classification, with all variables clean and complete, ensuring a robust base for model training and evaluation.

```python
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)


import os
print(os.listdir("../input"))
```

```
['Bank_Personal_Loan_Modelling.xlsx']
```

```python
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

file= pd.ExcelFile('../input/Bank_Personal_Loan_Modelling.xlsx')

description=pd.read_excel(file, 'Description')
df=pd.read_excel(file, 'Data')

description.head(10)
```



	ID	Age	Experie	Income	ZIP Co	Family	CCAvg	Educat	Mortga	Personal Loan	Securities Account	CD Acco	Online	CreditCard
11	10	34	9	180	93023	1	8.90	3	0	1	0	0	0	0
18	17	38	14	130	95010	4	4.70	3	134	1	0	0	0	0
20	19	46	21	193	91604	2	8.10	3	0	1	0	0	0	0
31	30	38	13	119	94104	1	3.30	2	0	1	0	1	1	1
40	39	42	18	141	94114	3	5.00	3	0	1	1	1	1	0
44	43	32	7	132	90019	4	1.10	2	412	1	0	0	1	0
49	48	37	12	194	91380	4	0.20	3	211	1	1	1	1	1
55	54	50	26	190	90245	3	2.10	3	240	1	0	0	1	0
59	58	56	31	131	95616	2	1.20	3	0	1	0	0	0	0
77	76	31	7	135	94901	4	3.80	2	0	1	0	1	1	1
80	79	54	30	133	93305	2	2.60	3	0	1	0	0	0	0
92	91	55	30	118	90277	4	5.60	2	0	1	0	0	1	0
133	132	58	34	149	93720	4	7.20	2	0	1	0	1	1	1
153	152	26	0	132	92834	3	6.50	3	0	1	0	0	0	1
162	161	29	0	134	95819	4	6.50	3	0	1	0	0	0	0
176	175	42	17	168	95503	2	7.90	2	0	1	0	0	1	0
185	184	29	3	148	92173	3	4.10	1	0	1	0	0	1	0
189	188	46	21	159	94305	3	1.90	3	315	1	0	0	1	0
201	200	36	11	158	92152	1	5.10	3	0	1	0	1	1	1
211	210	64	39	172	94707	4	3.10	1	282	1	0	1	1	1
245	244	65	39	170	90095	3	7.90	3	99	1	0	1	1	0
249	248	53	29	120	92626	4	2.70	2	111	1	1	1	1	0
253	252	54	28	170	92182	2	6.20	2	325	1	0	0	1	0
256	255	65	41	134	91942	3	3.90	3	121	1	0	0	1	0
263	262	42	16	111	93106	2	1.20	3	251	1	0	0	1	0

Using Decision Tree algorithm to predict the nature of loan acceptance

```python
df.columns
```

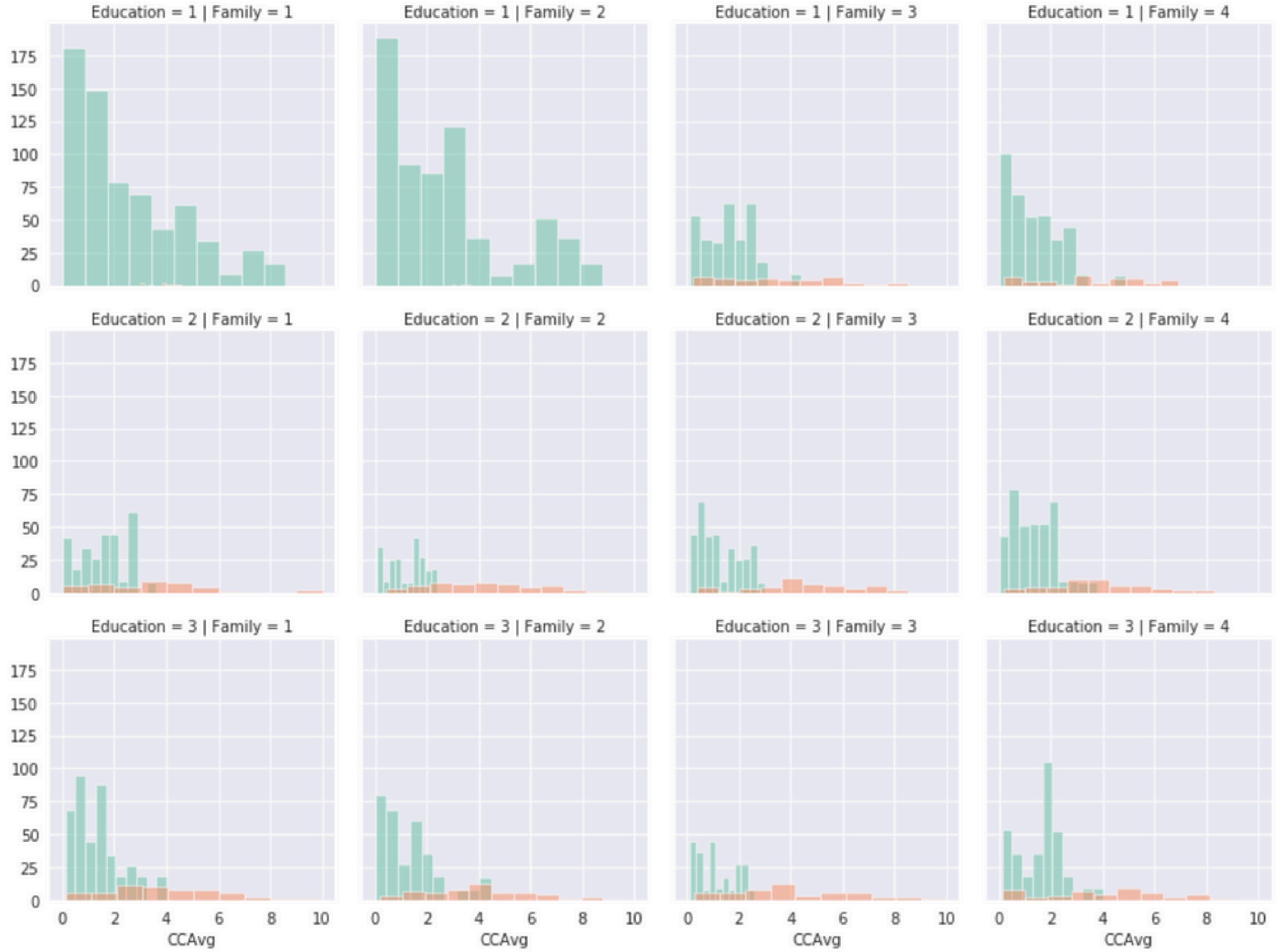
```
Index(['ID', 'Age', 'Experience', 'Income', 'ZIP Code', 'Family',
'CCAvg',
       'Education', 'Mortgage', 'Personal Loan', 'Securities Account',
       'CD Account', 'Online', 'CreditCard'],
      dtype='object')
```

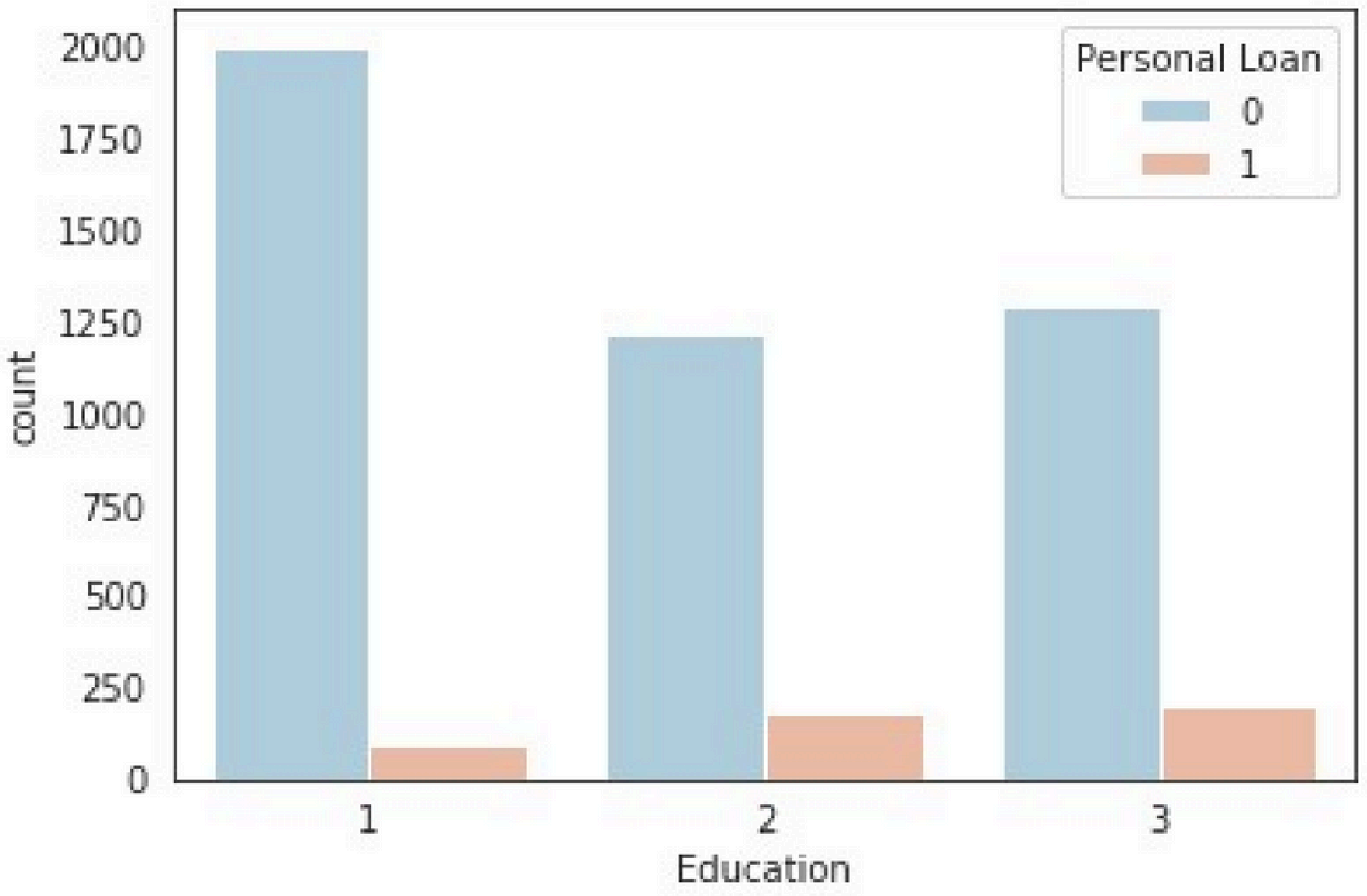
```python
X=pd.DataFrame(columns=['Age','Experience','Income','Family','CCAvg','
Education','Mortgage','Securities Account','CD
Account','CreditCard','Online'],data=df)

y=df['Personal Loan']

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test= train_test_split(X,y)

from sklearn.tree import DecisionTreeClassifier

dtree= DecisionTreeClassifier(max_leaf_nodes=3)

dtree.fit(X_train,y_train)
```

```
DecisionTreeClassifier(class_weight=None, criterion='gini',
max_depth=None,
            max_features=None, max_leaf_nodes=3,
min_impurity_decrease=0.0,
            min_impurity_split=None, min_samples_leaf=1,
            min_samples_split=2, min_weight_fraction_leaf=0.0,
            presort=False, random_state=None, splitter='best')
```

```python
predictions= dtree.predict(X_test)

from sklearn.metrics import classification_report, confusion_matrix

print(classification_report(y_test,predictions))
```

```
              precision    recall  f1-score   support

           0       0.96      1.00      0.98      1126
           1       0.99      0.63      0.77       124

   micro avg       0.96      0.96      0.96      1250
```






```
sns.set_style('white')
sns.countplot(data=df,x='Education',hue='Personal Loan',palette='RdBu_r')
<matplotlib.axes._subplots.AxesSubplot at 0x7826eff1cd68>
```

Dataset Snapshot & Target Variable

Key Features

The dataset includes features like Age, Experience, Income, Family, CCAvg, Education, Mortgage, Securities Account, CD Account, Online, and CreditCard, providing a comprehensive view of customer profiles.

1

Target Variable

The binary target variable 'Personal Loan' indicates whether a customer accepted the loan offer in the last campaign, with a 9:1 class imbalance reflecting real-world acceptance rates.

2

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Data Exploration Insights

Numeric Variables Descriptive Summary

Average Customer Profile

Customers average 45 years old, with 11 years of professional experience and an annual income of $71,000. Monthly credit-card spending averages $1,800, and mortgage balances average $56,000.

Data Completeness

The dataset has no missing values, and standard deviations show moderate spread, accelerating preprocessing steps and ensuring reliable model inputs.

Skewed Loan Acceptance

The skewed loan-acceptance rate highlights the need for precision-oriented metrics, as the model must accurately identify the minority class of acceptors.

Preprocessing Efficiency

The absence of nulls or extreme outliers simplifies preprocessing, allowing us to focus on feature engineering and model tuning for optimal performance.

Education & Family Size Influence

Education and Family Impact

Loan acceptance increases with education level, with professionals having the highest acceptance rates. Small families among undergraduates rarely apply for loans, indicating life-stage and credential effects on loan decisions.

Product Usage & Liquidity Signals

Liquidity Needs

Loan acceptors tend to have larger mortgages and higher credit-card averages, signaling liquidity needs. This insight helps refine risk scoring and identify cross-sell opportunities.

Securities Account Ownership

Very few loan applicants own securities accounts, suggesting that this product is less relevant to the target market for personal loans.

Mixed Product Adoption

Online and credit-card adoption is mixed across classes, indicating diverse customer preferences and potential for targeted marketing strategies.

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Model Development

Decision Tree Configuration Choices

Model Selection

We choose the DecisionTreeClassifier from scikit-learn with Gini impurity and limit max_leaf_nodes to three for transparency and interpretability, ensuring a simple model that stakeholders can understand.

Data Splitting

An 80-20 train-test split provides 4,000 training records and 1,250 test records, balancing robust evaluation with a straightforward, stakeholder-friendly tree structure.

Training & Prediction Workflow Steps

1

Training Efficiency

Training completes in milliseconds on the 4,000-record subset, producing a three-leaf decision tree that captures key patterns in the data.

2

Prediction Generation

The fitted model outputs acceptance probabilities for the 1,250 hold-out customers, generating binary predictions for performance assessment and marketing list generation.

3

Recursive Partitioning

The algorithm recursively partitions the feature space, selecting splits that maximize class purity, ensuring accurate and interpretable decision boundaries.

4

Model Deployment

The model is ready for deployment, providing actionable insights for targeted marketing campaigns and personalized loan offers to high-value prospects.

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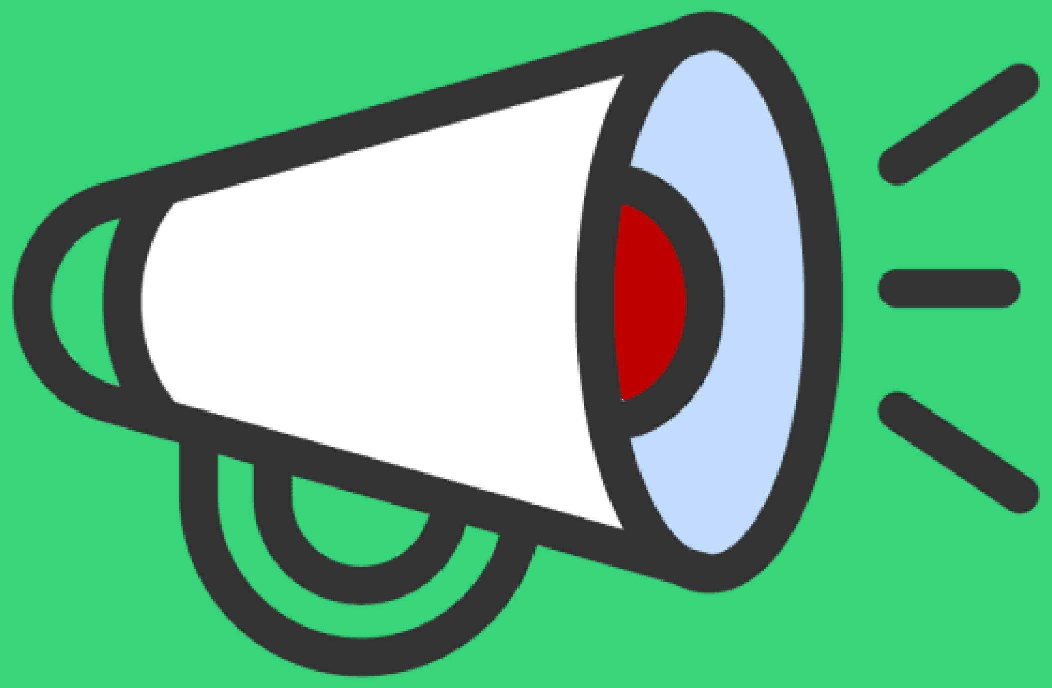
Model Evaluation

Classification Report Performance

Overall Accuracy

The model achieves an overall accuracy of 96%, with Class 0 precision at 96% and perfect recall, reliably identifying non-acceptors and ensuring efficient campaign targeting.



Confusion Matrix Numerical Breakdown

True Negatives

Out of 1,250 test cases, the model correctly identifies 1,125 true negatives, minimizing false positives and ensuring campaign resources are well-directed.

True Positives

The model captures 78 true positives, representing high-value leads worth pursuing with personalized offers and relationship-manager outreach.

False Negatives

Only 46 positives are misclassified as negative, indicating room for improvement in recall while maintaining high precision for acceptors.

Learned Tree Logic Revealed

1

Income Split

The tree opens with an income split, directing high-earners toward acceptance. This initial split captures a key driver of loan acceptance, aligning with business intuition.

2

Subsequent Splits

Subsequent splits on mortgage and education produce three leaves: one pure rejection node and two mixed nodes leaning toward acceptance, providing a clear decision path.

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